

EyeWithShut: Exploring Closed Eye Features to Estimate Eye Position

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ABSTRACT

Smart glasses enabling eyes-only input are growing in popularity. The underlying input technique powering the majority of interactions in this modality is dwell-based selection. However, this approach suffers from the Midas touch: the unintentional triggering of input during exploratory gaze. We suggest that multimodal input based on eye motions performed while the eyes are shut can trigger selections and overcome this problem without requiring additional input devices or modalities. To explore this idea, this paper captures a dataset of labeled closed-eye images corresponding to eye motions in various directions from nine participants. This was achieved by recording images from a binocular eye-tracker while participants performed standard eye-targeting tasks with one eye closed: the image from the open eye serves as ground truth for the closed eye's location. To understand the scope of closed-eye input, we explore this data according to three dimensions: the distance (near/far), width (narrow/broad), and direction (horizontal/vertical) of single captures of eye positions. Our results indicate that horizontal and vertical eye positions can be accurately recovered from static closed-eye images, but only after relatively large angular eye movements.

CCS CONCEPTS

• Human-centered computing → Interaction techniques.

KEYWORDS

Eye-tracking, Input technique

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1 INTRODUCTION

The emergence of heads-up computing based on smart glasses and mixed reality (MR) technology is allowing users to engage with their surroundings and move within space while they operate their devices. Eye-only interactions are being used in such scenarios

due to their convenience and the fact that they do not require users to carry external devices (e.g., hand-held controllers) or that they make large-scale gestures (e.g., as in hand-tracking-based input). Indeed, the recent release of Apple's Vision Pro showcased the power and versatility of interaction techniques that can be supported by gaze input [21]. However, eyes-only interaction is well known to suffer from problems of unintentional input, an issue known as the Midas touch[11]. This refers to the fact that, if eye gaze is used alone, inputs can occur unintentionally as users visually explore the environment: intentional selection can be confounded with simply looking around or in response to a particular trigger eye movement (e.g., an eye-swipe). In complex real-world environments where eye movements are continuous and highly diverse, such unintentional inputs will likely be frequent.

Closed eye input has been recognized as one of the multimodal channels for eyes-only interaction that can be employed to overcome the Midas Touch [1]. Essentially, a user closes one or both of their eyes while performing gaze input to specify some aspect of the interaction. This has the advantage of not requiring additional sensors or separate input modalities: everything is still controlled via the eye channel[1, 23]. Further, this approach can enhance traditional gaze-based interactions to achieve a richer and more intuitive experience. For example, input based on closing one or both eyes has been proposed for tasks such as password entry and gaming to create more discrete or immersive experiences [5, 22]. However, from a technical point of view, closed-eye input represents a significant challenge. When the eyes are closed, traditional eye-tracking methods cannot capture gaze direction. These strategies typically focus on the positions before and after a blink. However, such approaches do not directly capture any potential eye motion during the blink. We argue that in order to fully explore the potential of this input modality, we need to better understand the ability of eye-tracking systems to recover closed-eye locations from closed-eye images. This will, in turn, enable a richer set of input techniques.

Accordingly, this paper explores the viability of using features captured from closed-eye images to estimate gaze direction. To do this, we captured real eye images from the left and right eyes of nine participants as they performed target-tracking tasks with one eye shut. Our dataset includes approximately one million binocular eye images, and the data from the opened eye was used as ground truth to train Convolutional Neural Networks (CNN) classifiers on images from the closed eye. We explore the accuracy with which gaze position can be recovered with respect to key variables: the distance of the target area (near/far), the width of the target area (narrow/broad), and the direction of its displacement (horizontal/vertical). We report mean accuracies of between 85.16% and 86.40% for far target areas of both widths and directions.

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2 PRIOR WORK

2.1 Eye Input Technique

Current gaze input techniques enable accurate position-based interaction with content using just the eyes: we simply look at what we want. However, this paradigm suffers from the problem of unintentional input: the Midas Touch. Basically, as this paradigm cannot inherently distinguish between looking to visually explore an environment and looking to select some target [11], it suffers from frequent inadvertent target activations. To resolve this issue, eyes-based interaction is often used in conjunction with time- or motion-based techniques such as dwell [10, 17, 18, 25] or pursuits[3, 28] that can be used to explicitly separate exploratory and targeting motions. Additional approaches have also used eye gestures in the form of intentionally initiated saccades [8], eyelid gestures based on simple blinking patterns [12, 16] or squinting [30]. The work in this paper seeks to supplement these techniques by exploring the potential to recover positional information from closed-eye images.

2.2 Multimodal with Eye

Multimodal input techniques can overcome the unintentional input problem in eyes-only interaction. By using an additional sensor to expand the input channel, various modalities can be integrated with eye-gaze. For instance, Wang suggested combining touch and blink using smartwatches and smartglasses [29]. Similarly, Hild proposed combining physical touch buttons and gaze using the hand and foot [9]. Ludwig created hands-free interaction that combines control using head rotation and eye motion [26]. Gemicioğlu detected the tongue inside the mouth using the inertial measurement units of commercially available virtual reality headsets[7]. However, despite this promise, integrating and using additional sensors to detect input from channels other than the eyes can impair user experience and reduce usability. For example, such systems may require different computing batteries (e.g., they are physically separate devices) and increase the user’s mental load due to difficulties integrating gestures across different input channels. To enable multimodal input using only eye cameras, some prior work has suggested using additional gaze information, such as eye closure. For example, Arigenis combined winking and gaze for eye input techniques (e.g., scrolling, zooming) [24]. However, they note that with one eye closed, gaze input is slower, less accurate, and more demanding than that using both eyes.

2.3 Closed-Eye Estimation

Numerous approaches for mobile eye tracking have been explored in the past [6, 27]. One of the earliest studies to combine neural networks, artificial pictures, and near-eye images for gaze tracking was conducted by Balujal [2]. Indeed, CNNs have been demonstrated to be a promising and effective method for general gaze estimation [13, 15, 31]. Accordingly, we opted to apply this method in our closed-eye study. Finally, in work closely related to this paper and in order to identify the direction of closed-eye motion, Findling used optical flow calculations to create a machine learning model that recognizes cyclical closed-eye gestures with 4 directions: up, down, left, and right[4]. In this study, we move beyond this work

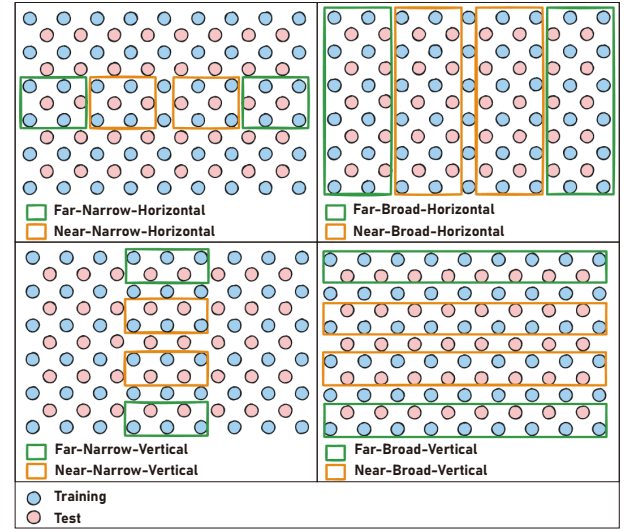


Figure 1: Illustration of the gaze areas used in the study. Horizontal (top images) and vertical (bottom images) areas relate to the angle of the target area with respect to the neutral forward gaze vector at the center of the grid. The width of the target area varies between narrow (e.g., left images) and broad (right images), reflecting how tightly the two directions are defined (e.g., broad areas encompass any movement in the defined direction, whereas narrow ones require only limited deviation on the orthogonal axis). Finally, the distance of the target areas varies between near (orange boxes in all images) and far (green boxes) from the forward gaze vector.

by seeking to analyze only single images and identify which fundamental characteristics of static closed-eye images influence the accuracy of the predictions of directions.

3 METHODS

3.1 Study Design

This study followed gaze estimation approaches and conducted data collection with one eye open and the other closed: standard tracking of the open eye was used to verify and label the location of the closed eye [14, 27]. Two interleaved target grids were employed: one with dimensions of 9x6, which served as the training data, and the other with dimensions of 8x5, which was used for testing (see Figure 1). Each trial in the data collection study involved showing a randomly rotated "E" at a point selected from either the testing or training grids. Participants’ task was to look at the target (with either one eye closed, or both open), then close one eye and use the keyboard to report the correct orientation of the "E". In all trials, the "E" was 4mm top-to-bottom and white displayed on a black background. After reporting, participants were then instructed to fixate on the displayed letter (with one eye still closed) continuously until it disappeared, a period of 1.5 seconds. Participants conducted six sessions (three times with the left and right eyes closed) each consisting of the "E" shown at 94 random grid points. To reduce the

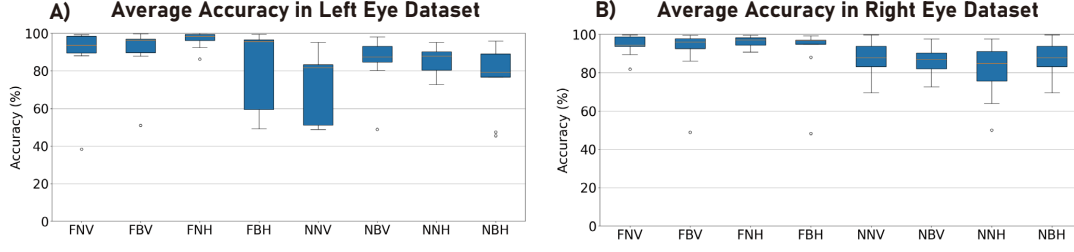


Figure 2: Box plots of average accuracy in 9 participants on left eye (A) and right eye (B) with 3 variable combinations. In detail, distance: far(F)/near(N), width: narrow(N)/broad(B), and direction: vertical(V)/horizontal(H)

impact of eye fatigue, participants were able to take a short break at any point during the data collection (between trials) and also took mandatory 2-minute breaks between sessions. If an incorrect response was reported (regarding the rotation of the "E"), the participant performed an additional attempt at the end of the session. We opted for this study design as targeting, when performed with one eye closed, results in lower accuracy, takes more time, and causes greater mental load than similar tasks with both eyes open [23].

3.2 Apparatus, Participants and Procedure

During the study, participants wore an eye tracker, the Pupil Neon [20], and placed their heads on a chin rest 40mm away from a 27-inch monitor with a pixel pitch of 0.16mm. The Pupil Neon has two IR eye cameras on nose pads recording both eyes at a resolution of 192 by 192 pixels and a temporal resolution of 200 Hz. Eye videos are recorded using a binocular pair of infrared cameras and matching 850nm infrared illuminator LEDs. Data are logged to a Motorola Edge 40 Pro smartphone which was connected to a Pupil Neon and the monitor. The display has a resolution of 1920 x 1080, covering a field of view of 44.8 degrees vertically and 72 degrees horizontally. To align one eye with the center of the display, we measured each participant's interpupillary distance (PD) and adjusted the display position to the left or right, as appropriate for each condition, by half of the PD.

Nine participants took part in this experiment. Six were males, three were female, and they had a mean age of 21.73 (SD 2.49). They were recruited from the local university via flyers. Three wore mascara, and two wore contact lenses. The mean PD was 6.66 cm, with a standard deviation of 0.54 cm. The participants were invited to a quiet room where no one was present except the researcher to minimize distractions. It was manually verified that they were capable of winking. After an explanation of the experimental protocols, participants who opted to continue with the study signed a consent form and completed a demographic survey. They sat on a chair in the experimental room and we adjusted the height of the chair and chin rest for comfort. We then adjusted the height and location of the display to match the height and horizontal position of their eyes. Participants then wore the eye trackers and practiced the study with each eye for a total of four minutes. All participants reported clearly viewing the letter "E" on the grid points used in the study. Afterward, a 2-minute break was provided. If performance in the practice was not sufficient, or they felt uncertain about the protocol, additional practice was conducted. Participants then began

the first session with their right eye open and their left eye closed. In each session, they alternated between opening and closing the left and right eyes. One session lasted 4 to 6 minutes. To account for eye fatigue, a 2-3 minute break was provided between sessions. In total, the experiment lasted approximately 1 hour, and we compensated the participants with 10\$ for their participation. The local institutional review board (IRB) approved the study protocol.

3.3 Data Explanation

To avoid capturing any initial or trailing eye motions, we recorded only the central one-second from the one-and-a-half seconds the participant gazed at the target in each trial. In addition, since winking eyes move even when closed, the dataset was checked and cleaned. Specifically, we used the ground truth image of the open eye as a label for the location of the closed eye. We utilized Pupil Neon blink detection to simply remove data in trials when the open eye blinked unintentionally [19]. In total, we collected approximately 1 million closed-eye images (200 FPS * 94 trials * 6 sessions * 9 participants) in our study.

3.3.1 Dataset Preparation. The position of closed eyes cannot be judged as accurately as with open eyes because there are fewer features, such as eyelid, eyelash, and skin glint positions, to serve as markers. Therefore, rather than try to recover the gaze location for each target location in our test set, we opted to explore performance in simpler binary tasks based on predicting the gaze target from different pairs of gaze target regions. These were selected to systematically vary according to the distance (near/far), width (narrow/broad), and direction (horizontal/vertical) of targets they encompass—see Figure 1 for full details. To explore how classification accuracy changes with these variables, we divided the dataset into eight different subsets according to each possible combination and constructed and trained classifiers to distinguish between two areas defined by each combination of the variables. For example, the two areas defined in the Far-Narrow-Horizontal combination entail positioning the eye in the relatively distant areas to the extreme left or right of the target grid and close to the horizontal axis.

3.4 Network Architecture

We trained the model per user by each combination of datasets and employed 5-fold-cross validation procedures. We used Convolutional Neural Network (CNN) architecture with eleven layers: two fully connected layers, a dropout layer, and five convolutional layers. The dimensions of the input layer are $192 \times 192 \times 1$. The

kernel size of 5×5 is used in each convolutional layer, along with a stride of 1 and a padding of 2, to guarantee that the output dimensions match the input dimensions. The output channels of the convolutional layers increase sequentially: 64, 128, 256, 512, and 1024. A ReLU activation function is applied to each convolutional layer to add non-linearity, and then a dropout layer with a dropout rate of 0.5 is applied to prevent overfitting. Furthermore, after every ReLU activation, max pooling is applied with a stride of 2 and a kernel size of 2×2 .

4 RESULTS

4.1 Classification Performance

4.1.1 Overall Description. Figure 2 shows the results for the left eye model. The Far-Narrow-Vertical condition had an accuracy of 88.85% (SD: 19.07), while the Far-Broad-Vertical condition had an accuracy of 89.22% (SD: 14.11). The Far-Narrow-Horizontal condition had a higher mean accuracy of 96.70% (SD: 4.53), while the Far-Broad-Horizontal condition had a mean accuracy of 82.00% (SD: 18.68). The Near-Narrow-Vertical condition had a mean accuracy of 73.31% (SD: 18.97), while the Near-Broad-Vertical condition had a mean accuracy of 84.80% (SD: 13.84). The Near-Narrow-Horizontal condition had an accuracy of 85.50% (SD: 7.86), while the Near-Broad-Horizontal condition had an accuracy of 77.37% (SD: 17.39).

In the case of the right eye model, the Far-Narrow-Horizontal condition exhibited a mean accuracy of 95.50% (SD: 2.96), whereas the Far-Narrow-Vertical condition demonstrated a mean accuracy of 94.75% (SD: 5.10). The Far-Broad-Horizontal condition demonstrated a mean accuracy of 90.19% (SD: 16.64), while the Far-Broad-Vertical condition had a mean accuracy of 90.54% (SD: 15.12). In contrast, the Near-Narrow-Horizontal condition exhibited a mean accuracy of 80.11% (SD: 15.82), while the Near-Narrow-Vertical condition demonstrated a mean accuracy of 84.16% (SD: 14.85). The Near-Broad-Horizontal condition exhibited a mean accuracy of 79.28% (SD: 13.21), while the Near-Broad-Vertical condition demonstrated a mean accuracy of 75.58% (SD: 22.42).

4.1.2 Distance. The analysis compared the accuracy between the far and near conditions for both the left and right eye models. For the left eye model, the far condition exhibited an average accuracy of 91.49% (SD: 15.18), whereas the near condition had a slightly lower average accuracy of 84.73% (SD: 14.55). An independent samples t-test indicated that this difference was statistically significant ($p = 0.020$). Similarly, for the right eye model, the far condition showed an average accuracy of 93.27% (SD: 4.10), while the near condition revealed an average accuracy of 81.95% (SD: 15.47). The independent samples t-test confirmed that this difference was also statistically significant ($p < 0.001$). These results suggest that the accuracy was significantly higher in the far condition compared to the near condition for both the left and right eye models.

4.1.3 Width. The results revealed there were no significant differences in accuracy between the narrow and broad conditions for either the left or right eye models. For example, for the left eye model, the narrow condition showed an average accuracy of 85.66% (SD: 17.67), while the broad condition revealed an average accuracy of 83.98% (SD: 16.80). These differences were not found to be statistically significant based on an independent samples t-test ($p = 0.488$).

For the right eye model, the narrow condition displayed an average accuracy of 86.77% (SD: 17.19), while the broad condition had an average accuracy of 85.96% (SD: 16.76), and this difference also was not statistically significant ($p = 0.806$). Therefore, we conclude there was no meaningful difference in accuracy between the narrow and broad conditions for either eye model.

4.1.4 Direction. The results demonstrated no significant differences in accuracy across direction conditions. For the left eye model, the vertical condition exhibited an average accuracy of 83.85% (SD: 20.33), while the horizontal condition had an average accuracy of 85.80% (SD: 13.57). The difference was not statistically significant according to an independent samples t-test ($p = 0.766$). Similarly, for the right eye model, the vertical condition exhibited an average accuracy of 84.80% (SD: 17.36), while the horizontal condition demonstrated an average accuracy of 87.91% (SD: 15.95). This difference was also not statistically significant ($p = 0.197$). These findings indicate no significant difference in accuracy between the vertical and horizontal conditions for either eye model.

5 DISCUSSION AND CONCLUSION

The far condition demonstrated significantly superior performance compared to the near condition ($p < 0.02$) for both left and right eye models. However, no significant differences in accuracy were observed between narrow and broad ($p > 0.48$) or vertical and horizontal conditions ($p > 0.19$). Prior work on closed-eye tracking indicated that an SVM model for closed-eye gestures using the optical flow can recognize one of four directions with an accuracy of between 92.9%-99.2% [4]. However, in specific cases, the models generated in this work show improved performance. As such, we believe the use of static images of closed eyes could enhance the performance of closed-eye tracking. In terms of limitations, we note that the advantages of having one eye open were not fully utilized. The open eye was used only to detect blinks for data filtering; gaze direction was not otherwise utilized. In the future, it would be beneficial to rely on the open eye as an indicator of correct task performance.

In conclusion, this work explored the impact of three variables on the performance of classifiers trained to recognize the position of a closed eye: distance, width, and direction. We captured 1M closed-eye images from both the left and right eyes of 9 participants as they performed target-tracking tasks with one eye shut. We trained a simple CNN classifier with different datasets that considered features and compared the accuracy. Only the distance variable led to statistically significant variations. Future work should develop closed-eye gesture estimation models utilizing static closed-eye images and test the usability of closed-eye input, such as input time and difficulty, in real-world scenarios (e.g., radial menu, walking while wearing the headset).

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